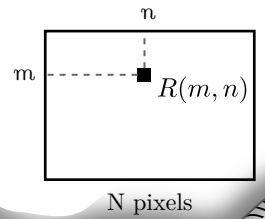
$$\tilde{R}(u,v) = \iint_{-\infty}^{+\infty} R(x,y)\, e^{-2i\pi(xu+yv)}\, dx\, dy$$

❶ ❷ ❸

Transformation de Fourier discrète

$$\tilde{R}(u,v) = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} R(m,n)\, e^{-2i\pi\left(\frac{mu}{M}+\frac{nv}{N}\right)}$$



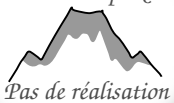
$R(x,y)$ dessin en niveau de gris

$R = 0 \rightarrow$ pixel noir

$R = 1 \rightarrow$ pixel blanc

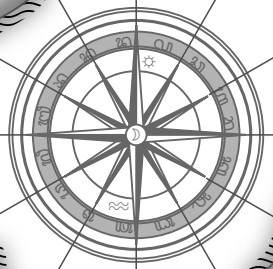
$e^{-2i\pi(xu+yv)}$

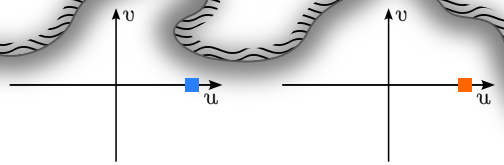
valeurs complexes



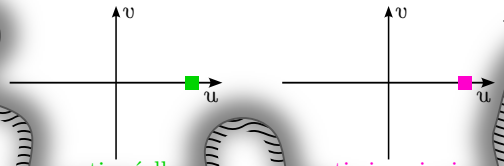
Pas de réalisation physiquement possible !

$e^{i\theta} = \cos\theta + i\sin\theta$





module argument $= \rho\, e^{i\varphi}$

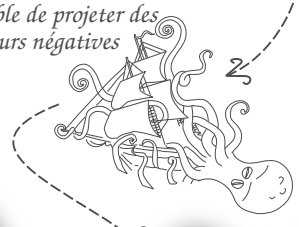


$\tilde{R}(u,v) = a + ib$

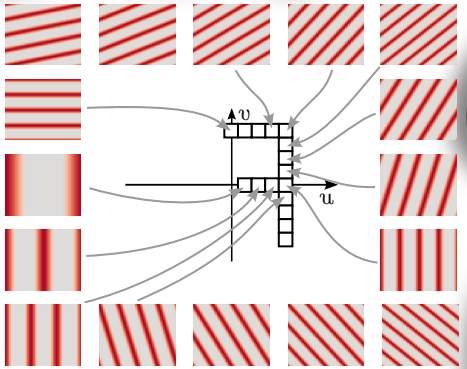
partie réelle partie imaginaire

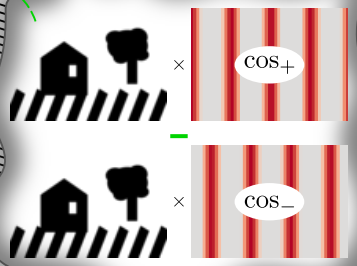
partie réelle *partie imaginaire*

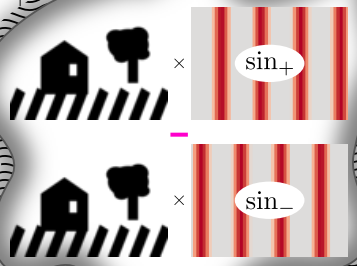
Impossible de projeter des valeurs négatives



Itération pour toutes les fréquences spatiales

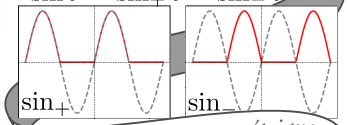






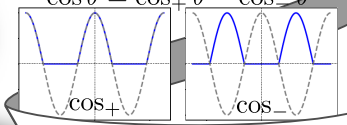
Décomposition

$\sin\theta = \sin_+\theta - \sin_-\theta$



des fonctions trigonométriques

$\cos\theta = \cos_+\theta - \cos_-\theta$



en fonctions positives